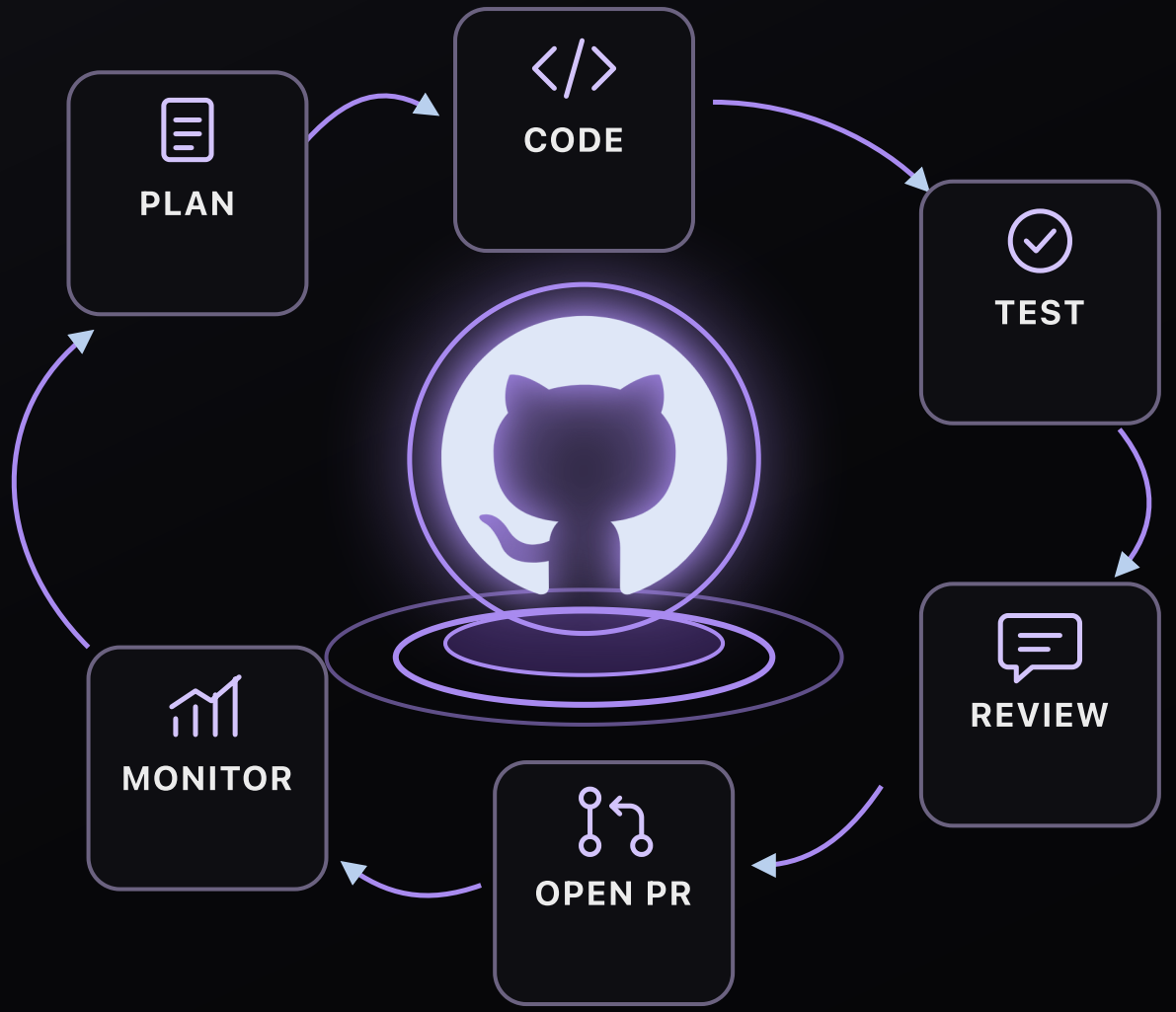
GitHub

GitHub Agentic Workflows

FROM AI THAT SUGGESTS TO AI THAT DOES

When did a page of documentation last mislead you?



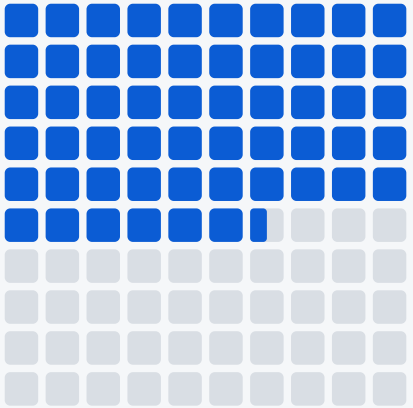
— 01 / THE PROBLEM

Outdated documentation already costs developers time

56.5%

of 12,374 developers say **outdated information** makes it hard to get the context they need.

In plain words: more than half



6,989 named it
56.5% of 12,374

5,385 did not
derived: 12,374 minus 6,989

1 square is about 1% of respondents

Industry survey (multi-select, about context in general), not our repositories · our own drift is not yet measured.

— 01B / THE PROBLEM

My earlier attempt solved part of the problem, and left a gap

What I built

tool name | what it did

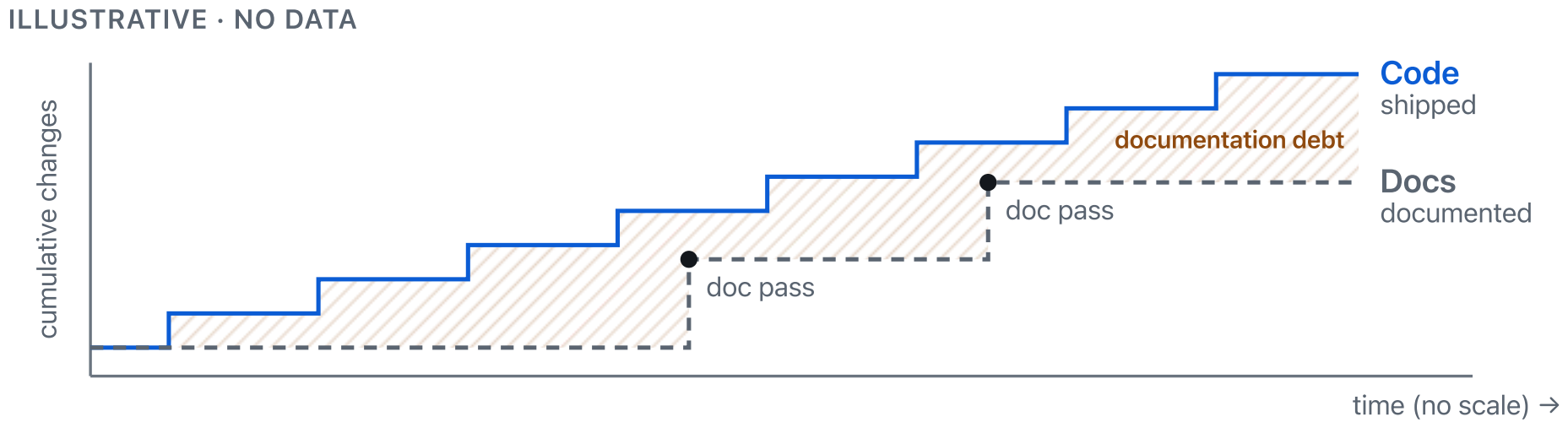
What it left undone

the remaining gap, one concrete instance

> Still blank: my own story. Nothing here is a claim until I supply it. Keep it to one tool, one limitation, one moment.

— 02 / THE MECHANISM

Documentation debt accrues between doc passes



> Every shipped change adds to it; each doc pass pays only some back.

Fig. 1 · Illustrative · no data · extends Cunningham's (1992) technical-debt metaphor to documentation.

— 03 / THE ASK

The ask is four weeks, one repository, and docs only

4 weeks

then a stop-or-go decision

1 repository

not yet chosen

Docs only

by instruction, reviewed in the diff

Draft PR

a proposed change, not yet merged; a person decides

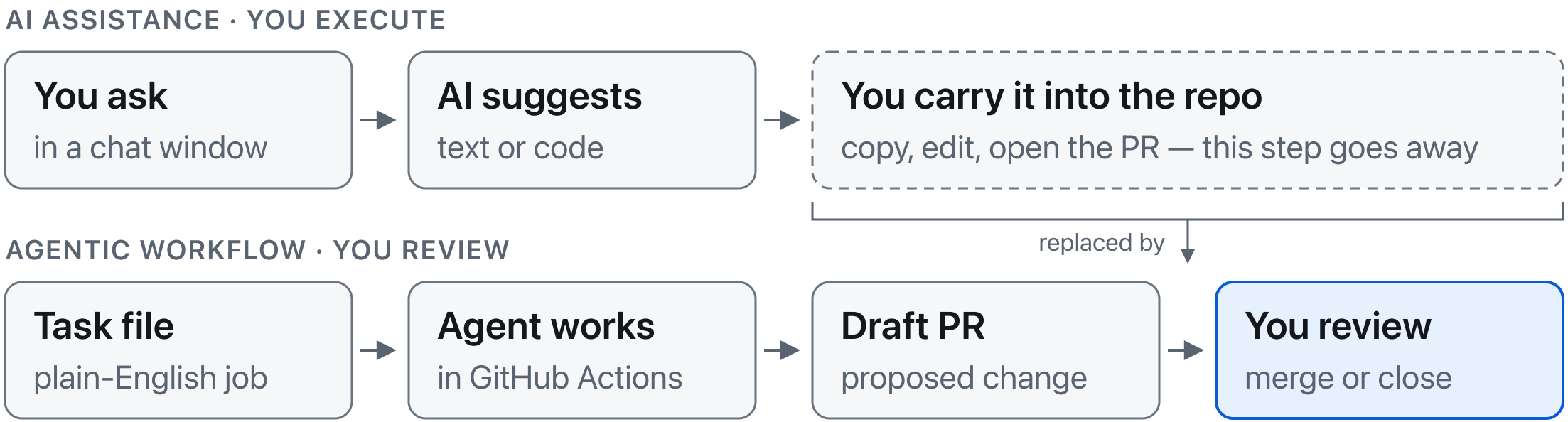
> Everything after this slide is evidence for or against that ask. The cost is unknown; the pilot measures it.

Still blank: **repository** **owner** **budget** **week-four thresholds**

This deck's proposal · public preview since 11 Jun 2026 · bounds: slide 14.

— 04 / THE SOLUTION

Agents shift the human role from execution to review



> The agent drafts, a person decides what merges: the work moves from writing to judging.

Fig. 2 · Illustrative, this deck's framing · `draft: true` is enforced PR policy · reviewer minutes not yet measured · Actions is GitHub's built-in automation.

Scripts check fixed rules; an agent is for the part that needs reading

First ask: can a fixed rule decide this?

Yes: use a script

Broken links · missing API entries · formatting

Cheaper, faster, predictable.

No, it needs reading: use an agent

“Does this paragraph still describe what the function does?”

Few fixed rules express that.

> Use the deterministic tool wherever one solves the task.

TODO · unverified: whether CI runs on the agent's own pull requests (the default GITHUB_TOKEN usually does not trigger workflows). Check the gh-aw docs first. Framing is this deck's, not GitHub's.

gh-aw docs: a third pillar beside CI and CD

— 06 / THE REAL EXAMPLE

A documentation agent's real pull request is eight added lines in one file

A workflow named “[docs] Update glossary - daily scan” opened it. It adds two glossary entries.

ADDED · docs/…/reference/glossary.md · +8 −0 · TRIMMED

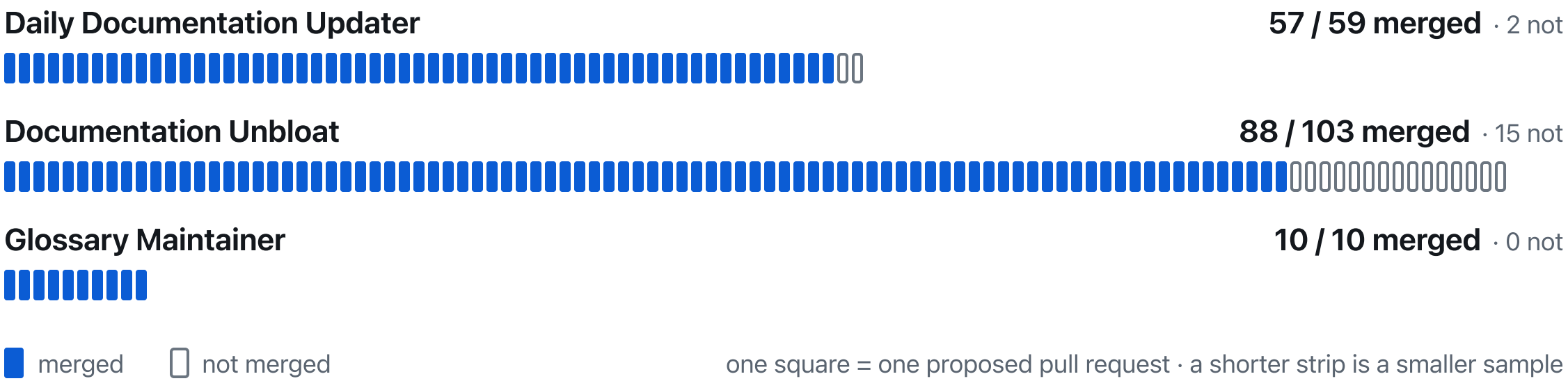
```
+ ### Work Queue (`tools.work-queue`)
+ A native, Git-backed queue for fair scheduling of
+ Work with immutable task dependency graphs. …
+
+ ### Claim (Work Queue)
+ Authorization for one attempt at a Work item in
+ a native work queue … See Work Queues.
```

> The unit of review is one small diff: a reviewer can judge it in minutes.

Public example · github/gh-aw #68654, merged 14 Oct 2026 · GitHub's own repository, not our pilot · text trimmed with …

— 07 / THE EVIDENCE

In three of GitHub's documentation workflows, most pull requests were merged, but not all



> Selected samples from other people’s repositories, not a forecast for our pilot.

Fig. 4 · GitHub report · 13 Jan 2026 · one square per proposed pull request, so strip length is sample size.

— 08 / THE RISK

It will be wrong sometimes, and that has to be survivable

Confidently wrong

It matches a page to code it misread.

LIMIT
A person reads the diff.

Too wide a diff

A one-page fix touches twenty files.

LIMIT
Draft PR, docs only by instruction.

Planted instructions

A pasted issue says “ignore your rules”.

LIMIT
No write access.

Spend, no merge

AI usage charges (credits) burn whether or not anything lands.

LIMIT
Spend caps: TODO

> Each has a limit we set, not trust in the model.

Failure modes this deck names · prompt injection is documented by GitHub; the rest follow from the task.

TODO · our own error rate is unmeasured — week one's staged preview is the first measurement.

Threat detection · Cost management

The limits live in a file you can read, not in the model

Think of it as a new colleague's onboarding sheet: when they work, what they can touch, what they can hand in. The block on top of the file is its settings (“frontmatter”); the text below is the job description.

```
---
on:                                    1
  push:
    branches: [main]
permissions: read-all                  2
safe-outputs:                          3
  create-pull-request:
    draft: true
---
# Update Docs                          4
Ensure every code-level change is
mirrored by clear, accurate, and
stylistically consistent documentation.
```

1. **Shift** `on:`
   Every push to main
2. **Access badge** `permissions`
   Reads the repository
3. **Outbox** `safe-outputs`
   Draft pull requests
4. **Job description** Markdown body
   Plain-English instructions

Real file, trimmed (description, labels, tools, network, comments cut; see next slide) · githubnext/agentics update-docs.md · PR policy

The same file also grants bash, web fetch and a network allowlist, so the other limits matter

```
network: defaults          # 1
tools:                     # 2
  github: {toolsets: [all]}
  web-fetch:
  bash: true
safe-outputs:
  create-pull-request:
    draft: true
    protected-files: fallback-to-issue   # 3
timeout-minutes: 15        # 4
```

1 · `network: defaults`
An allowlist of default hosts, not “no network”

2 · `tools`
Bash inside the Actions VM, web fetch, all GitHub toolsets

3 · `protected-files`
A guard on sensitive files; what it enforces is unconfirmed

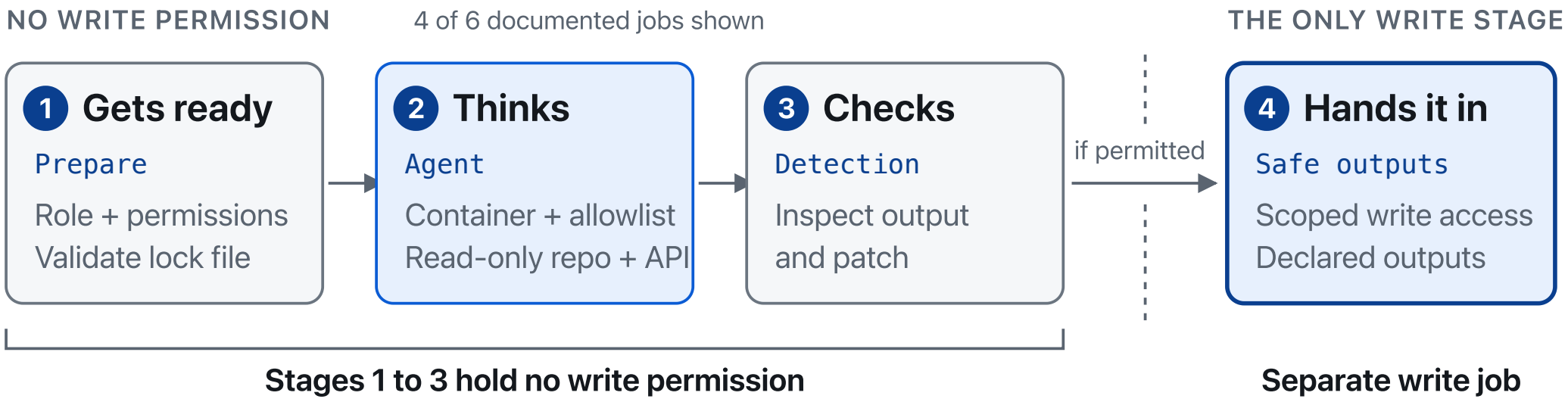
4 · `timeout-minutes`
Stops a run after 15 minutes

> TODO · confirm what `protected-files` enforces before calling “docs only” mechanical; pilot tool list is not chosen.

Real file, trimmed · githubnext/agentics workflows/update-docs.md

— 10 / THE RUN

The agent job has no write access; writes happen in a separate job



> A separate step applies outputs; a draft PR and human merge are the backstop.

Fig. 5 · Detection defaults to a caution, not a block · layers, not a guarantee · least privilege (Saltzer & Schroeder, 1975).

— 11 / COST

A cheap run nobody merges is still expensive

The right unit is cost per **accepted** change.

AI credits
from `gh aw logs`
NOT MEASURED

+

Actions minutes
from billing
NOT MEASURED

+

Reviewer time
from the pilot
NOT MEASURED

÷

Accepted changes
a reviewer said yes
NOT MEASURED

> Cost per accepted change: add the three, divide by what a reviewer accepted.

Fig. 6a · no run yet, so nothing here is measured · AIC = AI credit, 1 AIC ≈ $0.01 estimated · billing may differ.

TODO · Agree pilot limits; measure cost and review time · formula and symbols: Appendix 9.

Each week of the pilot has one job

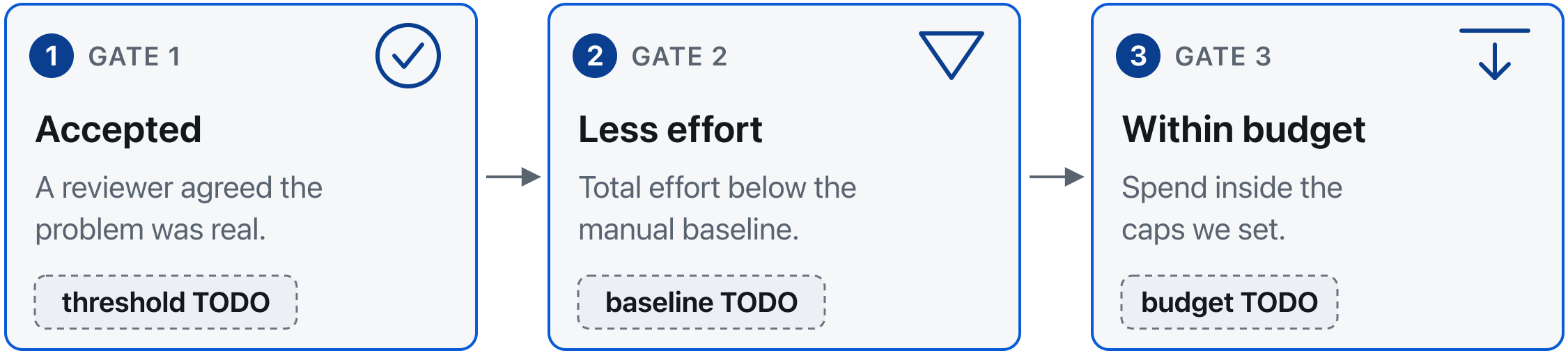


> Record accepted changes, reviewer effort and total cost together.

TODO · expected pull-request volume · a hard stop rule (for example, pause after N rejected drafts) · a setup-time estimate.

Fig. 6 · Week 1 runs with `staged: true`: writes are skipped, the run still costs.

Week four ends in a decision we agreed in advance



All three must pass; a miss is a real decision. Agreed before we start, so the result can tell us to stop.

TODO · all three thresholds, and the manual baseline they are measured against, need agreeing before week one.

Proposed criteria · this deck's recommendation, not an agreed plan.

The downside is bounded by five proposed limits

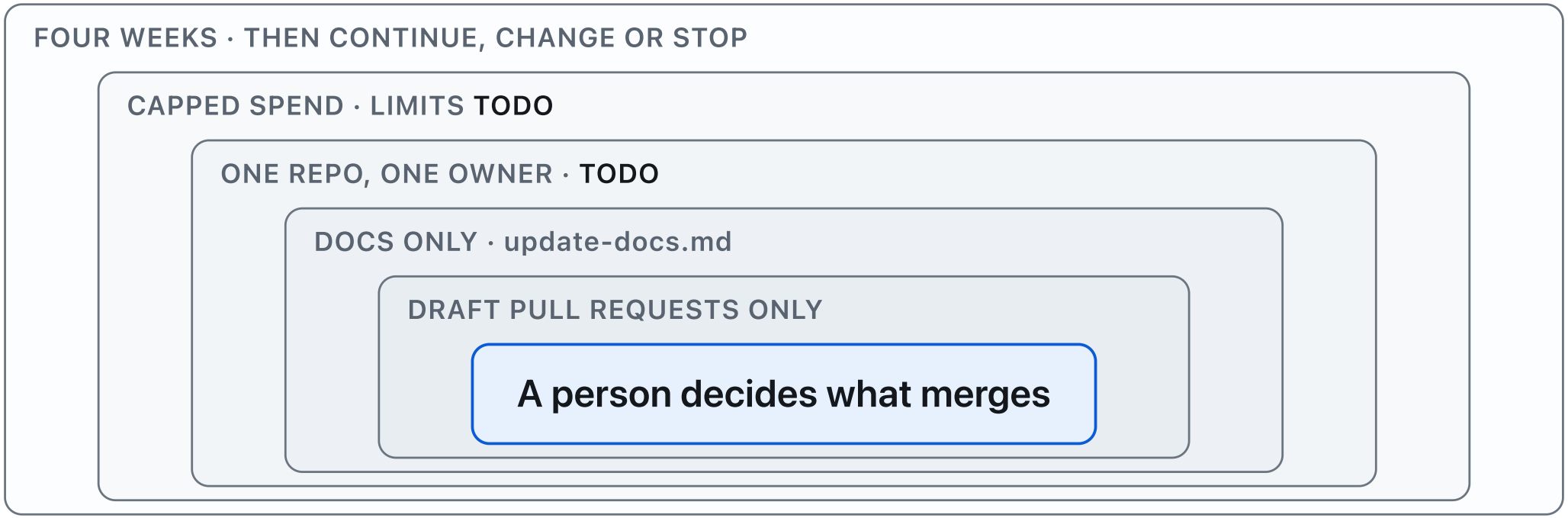


Fig. 7 · Proposed, not yet configured · spend, repository, owner TODO · “docs only” is by instruction and diff review; confirm `protected-files` before calling it enforced.

PR policy (draft: true)

Approve a four-week, docs-only pilot

Public preview since **11 Jun 2026**: features and defaults can change.

PROPOSED STARTING POINT

Candidate, a public third-party template, not our repository: `githubnext/agentics workflows/update-docs.md`

Every change is a draft pull request; a person decides what merges. Switch off at any time.

TO DECIDE BEFORE WEEK ONE

- [ ] Pilot repository + owner TODO
- [ ] Per-run + total pilot budget TODO
- [ ] Week-four thresholds + baseline TODO

Reversible by design · pause on unsafe output or a breached limit · proposal, not yet approved.

APPENDIX 1 / COMPILATION

Compilation turns that file into an Actions workflow

1 · Author

Markdown task
and frontmatter

2 · Compile

`gh aw compile`
Generated `.lock.yml`

3 · Publish

Commit both files
Run in GitHub Actions

> Compilation validates configuration and pins Actions to commit SHAs.

Agents add a third lane beside CI and CD

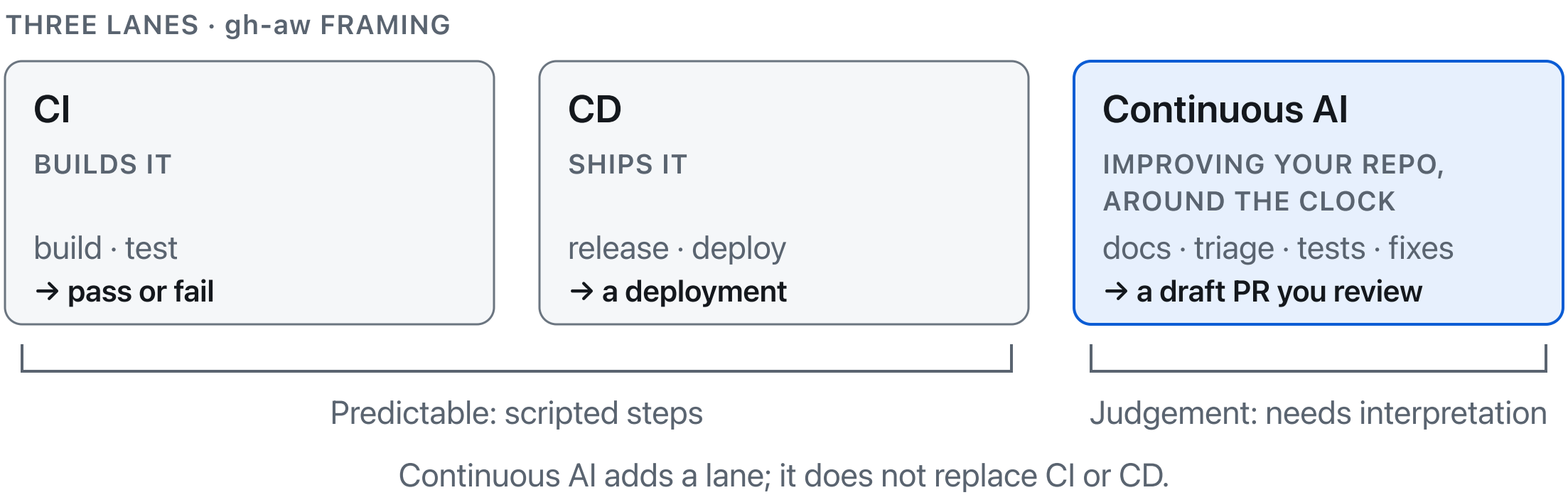


Fig. 8 · Lane captions quoted from the gh-aw docs · the predictable/judgement split is this deck's framing.

gh-aw docs: "A third pillar beside CI and CD"

Detection adds a review gate, not a safety guarantee

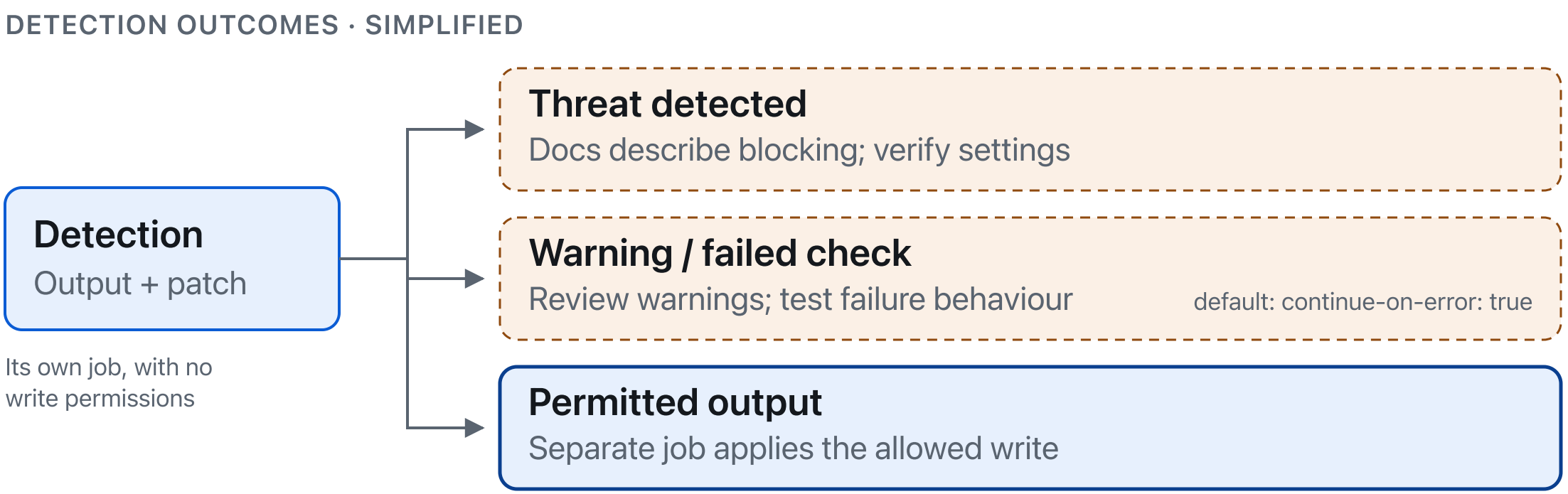


Fig. 9 · Failure behaviour is disputed in the docs; test it with explicit settings before relying on it.

APPENDIX 4 / DETAILS FOR QUESTIONS

Use agents where interpretation adds value

Good pilot candidate

Compare code with documentation.

Propose a small, reviewable correction.

Keep predictable work in CI

Build, test and release steps.

Use deterministic tools when they solve the task.

TODO · Agree which tasks stay with a person and our current autonomy level.

Start with one workflow from the example collection

Example collection

> TODO: repository URL
> TODO: workflow file
> TODO: recommended first task

Review before adding

Instructions and scope
Permissions and network
Outputs and budget

```
gh aw add-wizard owner/repo/workflow
```

Template, not runnable · replace it with the confirmed workflow reference.

APPENDIX 6 / DETAILS FOR QUESTIONS

A trial can run in a temporary private repository

Select

Choose the reviewed workflow file.

Trial

`gh aw trial <workflow>`
Temporary private repo by default.

Inspect

Read the results before choosing the next step.

> TrialOps is experimental.

Isolation changes where the workflow runs; it does not make the run free or risk-free.

Staged mode previews safe outputs before they are applied

```
safe-outputs:
  staged: true
  create-pull-request:
    draft: true
```

Compile and publish
Commit the Markdown and lock file.

Run and inspect
Read the Actions step summary.

Then decide
Enable draft PRs only after review.

> Staged safe-output writes are skipped; the workflow still runs.

The agent drafts; a person decides what merges

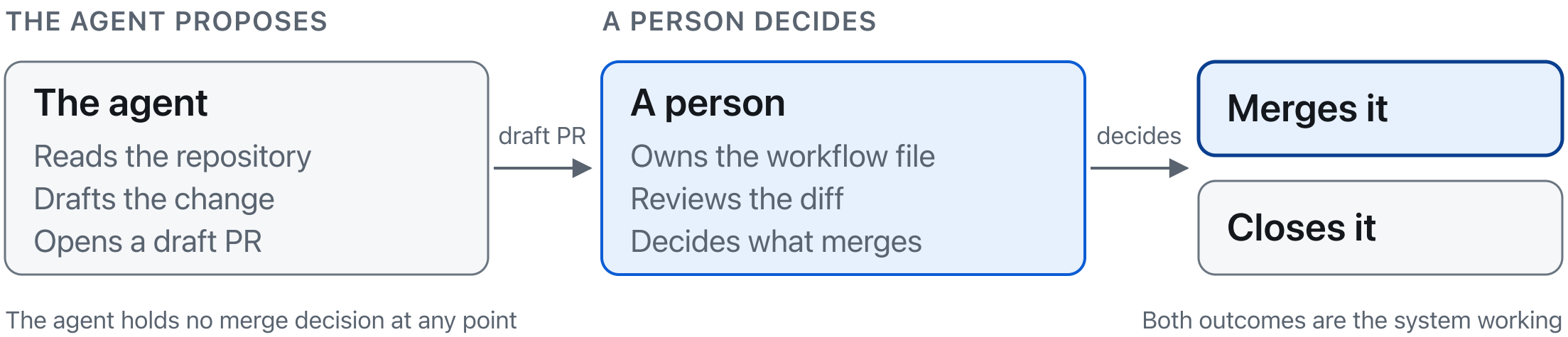


Fig. 10 · This deck's framing · `draft: true` is enforced pull-request creation policy · reviewer minutes are not yet measured.

Cost per accepted change is a formula the pilot fills in, not a number we have

$$C_{acc} = \frac{p_a A + p_m M + wR}{N_{acc}}$$

$$C_{man} = \frac{wR_{man}}{N_{man}}$$

Continue only if **$C_{acc} < C_{man}$**. The manual baseline is TODO.

Symbol	Meaning	Value
A	AI credits used (`gh aw logs`)	NOT MEASURED
p_a	price per credit	≈ $0.01, estimated
M	Actions minutes (billing)	NOT MEASURED
p_m	price per Actions minute	TODO
R	reviewer minutes	NOT MEASURED
w	reviewer cost per minute	TODO
N_{acc}	proposals merged	NOT MEASURED

This deck's formulation · no run yet, so nothing here is measured · AI credit (AIC) accounting is best-effort and billing may differ.

APPENDIX 10 / DETAILS FOR QUESTIONS

The technical claims are grounded in GitHub's documentation

Public preview since **11 Jun 2026**

Product and controls

Official changelog
Architecture and threat detection
PR policy and staged mode

Operating and measuring

CLI and schedule syntax
Cost management
Reported documentation results

Unresolved conflicts · daily AI-credit cap default · whether a failed detection run blocks safe outputs

Technical fact check: 17 Oct 2026; re-checked 18 Oct 2026. Sources are linked on the relevant slides and in speaker notes.

Public-preview changelog · Official docs